

Comparative Life Cycle Assessment of Battery and Hydrogen Energy Storage Systems for Connecting Renewable Energy to Grid

Course: Energy and Environment

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Abstract

This study presents a comparative life cycle assessment (LCA) of two key energy storage technologies, Battery Energy Storage Systems (BESS) and Hydrogen Energy Storage Systems (HESS), for integrating renewable energy into the electricity grid. Employing OpenLCA and the Ecoinvent 3.11 database, the analysis covers environmental impacts from cradle to grave using ReCiPe Midpoint (H), Cumulative Energy Demand (CED), and Environmental Prices methods. Results reveal that BESS has significantly lower greenhouse gas emissions (56 kg CO₂-eq/MWh vs. 137 kg CO₂-eq/MWh), lower energy demand (7,924 MJ-eq/MWh vs. 19,109 MJ-eq/MWh), and lower environmental costs (30.39 €/MWh vs. 75.31 €/MWh) than HESS. Material resource analysis highlights BESS's dependence on lithium and spodumene, while HESS is more reliant on cerium and platinum. Economic evaluation further confirms BESS's cost advantage, with a Levelized Cost of Storage (LCOS) of 116.11 €/MWh versus 259.81 €/MWh for HESS. The inclusion of environmental externalities further widens this gap. The findings underscore BESS as the more sustainable and economically viable option for short- to mid-term storage, whereas HESS may be justified for long-duration or seasonal applications, subject to technological advancements and cost reductions.

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1. Literature Review

The installation of variable renewable energy has expanded worldwide to help reach the goal of a sustainable society (IRENA, 2023). However, the intermittent nature of power generation from variable renewable energy increases the difficulty to guarantee a stable power supply. A huge amount of renewable power has to be curtailed without the proper control in the power grid (Vaziri Rad et al., 2023). Various flexibility measures have been proposed for managing power generated from variable renewable energy. Battery energy storage systems (BESS) and hydrogen energy storage systems (HESS) are among the key technologies addressing this challenge. Batteries are inherently energy-intensive products that would result in significant carbon emissions during their entire lives, although the development of BESS is partially triggered by the incentives to reduce the environmental impacts (Han et al., 2023). While green hydrogen is often promoted as a zero-carbon solution, its full life cycle environmental impact remains a subject of debate (Chelvam et al., 2024). Therefore, it is attracting worldwide interest to evaluate and compare BESS and HESS towards greener and more sustainable energy storage systems. In this chapter, the current study in this area is reviewed, helping to understand how this issue has been addressed and find the knowledge gaps.

1.1. Technology Background

To promote the penetration of variable renewable energy and avoid the limitation of spatial conditions caused by pumped hydro energy storage, the ideal energy storage types are batteries and hydrogen storage. This section summarizes the technological state of both systems.

1.1.1. Battery Energy Storage Systems (BESS)

Rechargeable batteries can charge and discharge repeatedly by converting electrical and chemical energy in both directions. Many types of batteries have been developed, with lithium-ion battery (LIB) making up most of the BESS, which account for 90% of the entire grid-connected industry (Badawi M et al., 2025). These batteries are leading technologies in the field because of their high energy and power density, low self-discharge, and long lifespan (Madhavi & Vairavasundaram, 2024). The LIB cathode is made of lithium metal oxide such as LiCoO_2 and LiFePO_4 , and the anode is made of carbon. The electrolyte is an organic-liquid solvent dissolving lithium salt, such as LiPF_6 (Mahmud et al., 2022).

1.1.2. Hydrogen Energy Storage Systems (HESS)

Hydrogen has the potential to become a relevant energy carrier for long-term and large-scale energy storage due to its low level of self-discharge, stackable capacity, and high energy density (Hassan et al., 2023). Hydrogen can be produced from fossil fuels, biomass resources, or water through processes such as pyrolysis and electrolysis (Zhang et al., 2016). In terms of storage, hydrogen can be stored in several ways, including compressed gas, cryo-compressed liquid, or with metal hydrides (Badawi M et al., 2025). Power generation using hydrogen includes fuel cells and hydrogen gas turbines. Fuel cells with hydrogen and oxygen are described as a cleaner alternative, as they do not go through a combustion process (Madhavi & Vairavasundaram, 2024).

Currently, the interest in utilizing the hydrogen storage integrated grid as energy storage is increasing. Several stationary projects based on hydrogen energy storage integrated grid system has been developed all over the world. Hydrogen allows the delivery of energy in higher capacities while lowering the transmission losses,

effectively manage large variations in electrical power and capitalizing on the surplus energy from renewable sources that otherwise be curtailed (Irham et al., 2024). A typical HESS system applied in this scenario contains electrolyzers, hydrogen tanks and fuel cells (Alonso et al., 2024).

In terms of water electrolysis, there are three principal electrolysis technologies, namely Alkaline Electrolyzers (AE), Proton Exchange Membrane Electrolyzers (PEME), and Solid Oxide Electrolyzers (SOE). Table 1.1 shows a comparison of key aspects considered for technology selection (Madhavi & Vairavasundaram, 2024). Among the fuel-cell technologies, polymer-electrolyte-membrane fuel cell (PEMFC) and solid-oxide fuel cell (SOFC) both have a relatively high technological maturity and large research-and-development activities. In general, the most critical materials in HESS are those that are used as catalysts for electrochemical reactions. This applies to AE, PEME and PEMFC, where the catalysts are generally based on platinum-group metals (PGMs). Due to a different mode of operation and higher temperatures, SOE's and SOFC's core components are mainly based on yttrium, cerium, lanthanum and other rare-earth elements (REEs), which are heavily dependent on imports in Europe (Mori et al., 2021).

Table 1.1: Comparison of Three Electrolysis Technologies

Aspects	AE	SOE	PEME
Operating Temperature	50–70 °C	500–900 °C	50–80 °C
Current Density (A/cm ²)	0–4	0–4	0–4
Response Time	Moderate	Moderate to Slow	Fast
System Complexity	Moderate	Complex	Moderate to Complex
Commercial Maturity	Mature	Emerging	Mature
Electrolyte	Alkaline solution	Solid ceramic oxide	Solid polymer membrane
Efficiency Range (LHV)	60–70%	>80%	70–80%
Cell Voltage (V)	0–2.5	0–2.5	0–2.5
Hydrogen Purity	High	High	High
Cost	Moderate	High	Moderate to High

1.1.3. Hybrid Integration of Batteries and Hydrogen

The hybridization of different energy storage options is a hot topic when considering storage possibilities in energy systems design due to the synergy of combining various technologies with complementary characteristics. Originally developed for fuel-cell electric mobility, the integration of batteries and hydrogen is now gaining significance in stationary Power-to-X (PtX) and Power-to-Power (PtP) applications within microgrids. Hybrid Energy Storage Systems (HESS) have demonstrated significant reductions in Levelized Cost of Energy (LCOE) and operational costs, along with improved efficiency and less component degradation. At the grid level, a combination of alkaline water electrolyzers and Proton Exchange Membrane (PEM) fuel cells, together with lithium-ion batteries, has shown cost competitiveness when compared to battery-only systems for storing excess energy from wind turbines. This suggests that hydrogen offers the viable capacity for long-duration storage and can overcome the seasonal storage limitations posed with Battery Energy Storage Systems (BESS). In addition, previous literature indicates that both hydrogen and battery storage facilities were somewhat capable of mitigating the managers concerns over price volatility and captured the potential of arbitrage within the combined use of hydrogen and battery storage systems (Alonso et al., 2024).

1.2. The State of Art from a LCA Perspective

Instead of simply comparing the operational performance and economic cost, the environmental sustainability of these energy storage systems can be better understood and evaluated by a comprehensive life cycle assessment (LCA) approach. Recent studies have focused on the life cycle performance of both BESS and HESS,

evaluating their production, use and end-of-life stages. Based on a review of recent LCA studies, this section summarizes how battery and hydrogen energy storage systems have been evaluated in terms of environmental impacts across different life cycle stages and application scenarios.

1.2.1. Battery Energy Storage Systems (BESS)

The life cycle environmental impacts of BESS have been widely analyzed in recent literatures. Yudhistira et al. (2022) provided insights into the cradle-to-grave environmental impacts of lithium-ion batteries compared to lead-acid batteries for grid-scale energy storage. This study addressed gaps in existing LCAs by focusing on end-of-life recycling and evaluates five environmental categories: climate change, acidification, fossil and mineral resource use, and particulate matter emissions. It highlighted that manufacturing processes, particularly cathode production, drive acidification and mineral resource impacts due to the extraction and processing of metals like nickel and cobalt. Similarly, Han et al. (2023) emphasized that material extraction and cell manufacturing dominate the life cycle greenhouse gas (GHG) emissions of BESS, while the use phase contributes minimally to the total environmental burden. Furthermore, Le et al. (2024) pointed out that end-of-life management for BESS remains challenging, as current recycling technologies are inefficient and recycling rates for critical minerals are low, potentially exacerbating environmental impacts if second-life applications are not broadly adopted. For BESS, the production phase remains the most environmentally impactful. The key mitigation strategies focus on improving material efficiency, scaling up recycling capabilities, and promoting second-life battery use.

1.2.2. Hydrogen Energy Storage Systems (HESS)

In contrast, HESS presents a different environmental profile, especially suited for long-duration and large-scale applications, while its environmental footprint hinges on metal-intensive renewable infrastructure and electrolyzer choice. Weidner et al. (2023) assessed the environmental impacts of large-scale hydrogen production technologies assuming different life cycle background inventories representing possible future economies with a higher degree of decarbonization. According to this study, the extraction and processing of metals for renewable energy infrastructure contributed the most to the environmental impacts of green hydrogen production, pointing to a need to improve metals recycling and mining practices. Zhao et al. (2020) performed a detail quantitative environmental impact assessment on the three commercially H₂O electrolysis technologies, namely SOE, PEME, and AE. Life cycle impact results from PEME stack were much higher than these from SOE and AE stacks, while electricity played a more important role in the life cycle impact of hydrogen production. Rangel et al. (2025) evaluated the life cycle impacts of green hydrogen using the True Cost of Hydrogen (TCOH) framework production and revealed that over 97% of the environmental cost of hydrogen (ECOH) is concentrated in four impact categories: Human Carcinogenic Toxicity (HCT), Fine Particulate Matter Formation (FPMF), Human Non-Carcinogenic Toxicity (HNCT), and Global Warming Potential (GWP). Moreover, Mori et al. (2021) compared the materials used in fuel-cell and hydrogen technologies using a novel methodological assessment and with a LCA approach. The results indicated that PEME electrolyzers had the highest environmental impacts, lower impacts were observed in PEMFC fuel cells and SOFC fuel cells, while the lowest impacts were seen in AE electrolyzers.

1.2.3. Comparative Analysis of BESS and HESS

Considering the comparative performance of BESS and HESS, recent studies reveal that the optimal choice between technologies is highly dependent on the storage duration, cycling frequency, and energy system context. Gandiglio et al. (2022) examined a hybrid energy storage system (HESS) combining solar, lithium-ion batteries, and hydrogen storage for an isolated community. The results showed an 89% reduction in greenhouse gas emissions compared to a diesel-only scenario, highlighting the environmental benefits of HESS despite some material and human toxicity impacts during manufacturing and transportation. Alonso et al. (2024) evaluated various storage configurations, from battery-only systems to hydrogen-only and hybrid systems. The findings showed that while battery systems are cost-effective in grid-connected scenarios due to their high efficiency, hydrogen is crucial for off-grid or weak-grid situations, offering deeper decarbonization and addressing seasonal imbalances. It also highlighted that hybrid systems (HESS) provide the best solution for off-grid operations. Le et al. (2024) assessed the environmental impacts of hybrid energy storage systems (BESS-HESS) across varying self-sufficiency ratios (SSR) for grid-connected buildings. The study suggested that while BESS is more efficient for short-term storage, HESS performs better for long-term or seasonal storage. However, it excluded end-of-life impacts and didn't consider the effects of recycling, limiting its conclusions. Yamaki et al. (2024) conducted a LCA comparing three energy storage systems, BESS, HESS, and Thermal Energy Storage (TES), for integrating large-scale wind energy into the grid. The study suggested that for large-scale energy storage installations, HESS or TES may be more suitable than BESS due to their lower GHG emissions and ARD, with the choice between HESS and TES depending on regional demands for hydrogen and heat.

1.2.4. Summary of LCA Aspects

In the reviewed studies, several life cycle aspects were systematically assessed for both BESS and HESS. A summary of these considerations is presented in Table 1.2.

Table 1.2: Summary of LCA Aspects

LCA Aspect	BESS	HESS
GHG Emissions	Dominated by battery production (e.g., lithium, cobalt, nickel) and mining activities . Lower emissions in operation.	Hydrogen production (electrolysis) is a key contributor to emissions but can be reduced with renewable energy for electrolysis.
Resource Depletion	Significant impact due to lithium, cobalt, and nickel extraction for batteries. Concerns over limited availability of these materials.	HESS involves substantial resource demand for electrolyzers, fuel cells, and storage tanks . Emphasis on improving recycling .
Energy Use	High energy consumption during battery manufacturing , especially for lithium-ion batteries. Lower energy use in the operational phase.	High energy consumption for hydrogen production , particularly when non-renewable energy is used. Efficient in large-scale systems.
Toxicity	Concerns over human and eco-toxic risks from chemicals used in battery production (e.g., mining, manufacturing processes).	Human toxicity and ecotoxicity associated with the production of electrolyzers and fuel cells .
End of Life Management	Battery recycling remains a challenge due to inefficiencies and low recycling rates for critical materials like lithium, cobalt.	Fuel cell and electrolyzer recycling still in early stages. Lack of established EoL systems .
Operational Efficiency	High round-trip efficiency : BESS is more efficient in storing and discharging energy, particularly for short-term applications.	Lower round-trip efficiency : Involves multiple processes (electrolysis, compression, fuel cell conversion), impacting energy efficiency.
System Integration	Well-suited for short-term storage (high-frequency cycling), optimal for daily to weekly energy fluctuations.	Better for long-duration storage and seasonal energy balancing . Offers synergy in hybrid systems with batteries.
Cost-effectiveness	BESS is typically cheaper for short-term storage due to mature technology and higher round-trip efficiency.	HESS is more cost-effective in low-grid availability or seasonal energy storage but requires high upfront costs for electrolyzers and fuel cells.
Scalability	Effective for small to medium-scale energy storage needs. Performance can degrade with increased storage capacity.	Highly scalable for large-scale, long-duration storage. Ideal for large installations with high seasonal demand.

Overall, while existing LCAs provide critical insights into the environmental profiles of BESS and HESS individually, direct and consistent comparative studies remain scarce. Furthermore, gaps exist in system-wide analysis that incorporates realistic operational strategies, integration with renewable energy profiles and future technology. Addressing these gaps is crucial to guide policymakers and stakeholders in making decisions about sustainable energy storage deployment.

2. Introduction and Objectives

In light of the current challenges, this study aims to conduct a systematic and comparative life cycle assessment of BESS and HESS under consistent assumptions and realistic system configurations. The scenarios are designed to reflect real-world applications in renewable energy integration, grid stability, and long-duration storage. By evaluating their sustainability and effectiveness in integrating renewable energy into the grid, it will further help to select the appropriate energy storage system. The project objectives are summarized as follows:

- 1) Assess the environmental impact of BESS and HESS over their full life cycle, from raw material extraction to decommissioning.
- 2) Compare greenhouse gas emissions, energy efficiency, resource consumption, and other related aspects of both storage technologies. Mainly the Life cycle Impact assessment will assess the Climate Change, Material Resources, Energy Demand, Environmental Price, as well as recycling and end of life considerations.
- 3) Evaluate the cost-effectiveness of energy storage systems over their operational lifespan by using the Levelized Cost of Storage (LCOS).

- 4) Provide policy and technology recommendations to support the deployment of sustainable energy storage systems in the energy transition.

3. Methodology

A life cycle assessment (LCA) approach has been considered to assess the environmental impact of BESS and HESS. The analysis is carried out using openLCA 2.4.0 software and Ecoinvent 3.11 database.

3.1. Energy Storage System Configuration

The technical data of the case study is based on a reference model of energy storage system by (Yamaki et al., 2024), which simulates and compares energy storage systems with batteries, H₂ storage and thermal energy system. The system configuration and energy flows are shown as Figure 3.1. Essentially, the same amount of electricity is generated by renewable energy. In the BESS system, LIBs are used to store the excess electricity and supply it to the grid if needed. In the HESS system, the excess electricity is converted to hydrogen through the AE electrolyzers, stored in the hydrogen tank, converted to electricity through the PEMFC fuel cells and supplied to the grid when needed. Energy curtailment is considered for both systems to simulate the condition that renewable energy needs to be curtailed when the storage system is fully charged. More energy is curtailed in HESS because the electrolyzer has a stricter working condition than the battery.

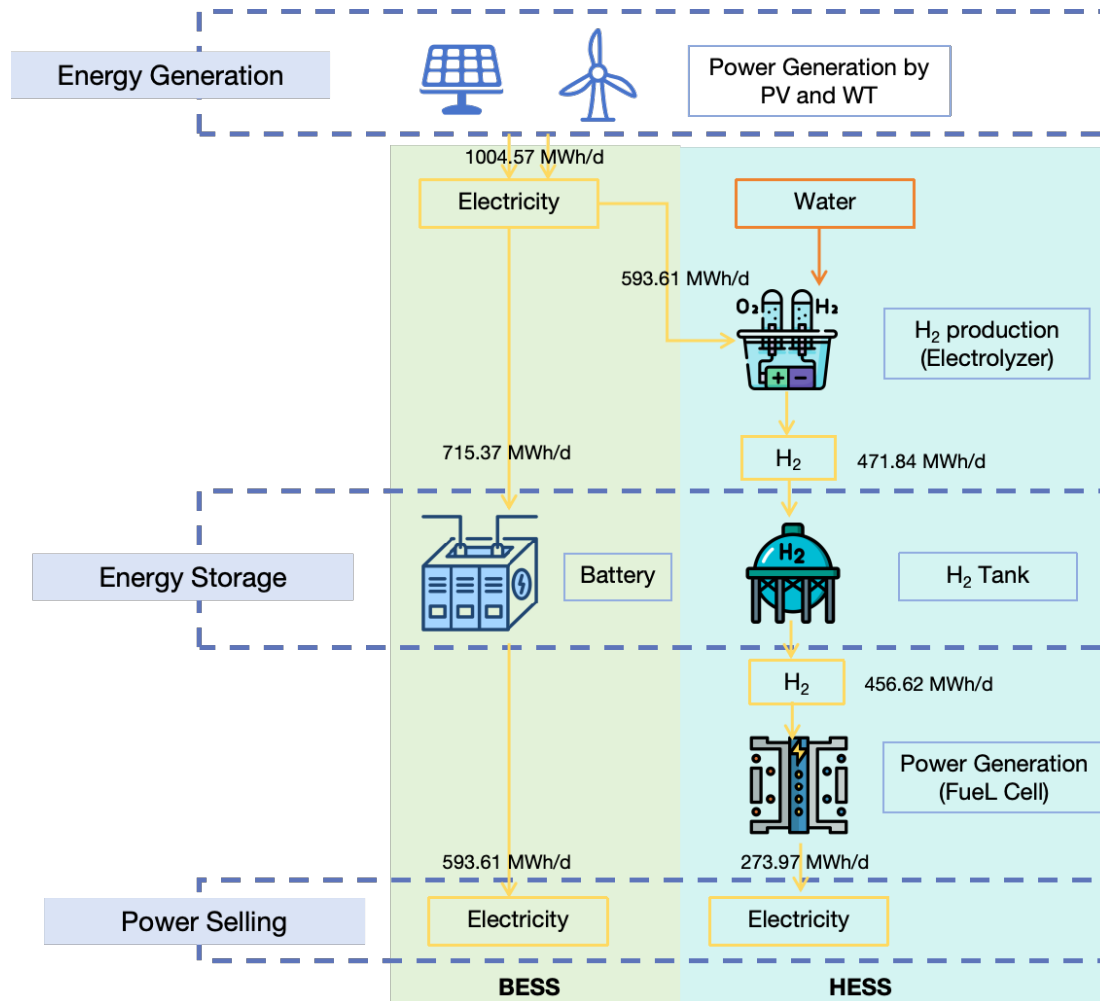


Figure 3.1: System Configuration and Energy Flows

Based on the energy flow of both BESS and HESS system in model and the basic industrial design rules, we design the BESS system (Table 3.1) and HESS system (Table 3.2) as follows. Project life is fixed as 20 years. Detailed calculations are provided in the appendix.

Table 3.1: Technical Data of BESS System

Battery Type	LIB-LEP
Full Cycle Duration (hours)	4
Daily Cycles	2
Battery Energy Capacity (MWh)	357.69
Battery Power Capacity (MW)	89.42
Lifetime (hours)	29200
Lifetime (years)	10

Table 3.2: Technical Data of HESS System

Electrolyzer	
Electrolyzer Type	AE
Daily Duration (hours/d)	8
Electrolyzer Power Capacity (MW)	74.20
H ₂ production (kg/day)	11872.15
Lifetime (hours)	58400
Lifetime (years)	20
Hydrogen Storage Tank	
Hydrogen Tank Type	Type III, 350 bar, compressed gas
Compressed Gas Volume (m ³ /day)	498.83
Volume of tank (m ³)	500.00
Lifetime (years)	10
Fuel Cell	
Fuel Cell Type	PEMFC
Daily Duration (hours/d)	8
Fuel Cell Power Capacity (MW)	34.25
Lifetime (hours)	29200
Lifetime (years)	10

3.2. System Boundary for LCA Study

This study aims to compare the cradle-to-grave environmental impact of BESS and HESS systems for connecting renewable energy to grid. The upstream renewable energy supply is not included in the scope of LCA assessment. Figure 3.2 provides the system boundary of the study. Specific data collection of transportation phase is excluded from the study due to a lack of location-specific data for supply sources. Modes, distances, and routing are often assumptions, leading to high uncertainty. This is supported by literatures that most of the relevant studies focus more on

manufacturing and operation stages (Baumann et al., 2017) (Bhandari et al., 2014). Instead, we choose "market for..." or "market group for..." flows in OpenLCA with the Ecoinvent database, where transportation from production to market region is already included.

The study case in Belgium (Alonso et al., 2024) is used as a reference geographical reference for the processes. If no processes with the geographical reference of Belgium is available, processes with the geographic reference Europe will be chosen. Otherwise, RoW (Rest of the World) of GLO (Global) will be chosen.

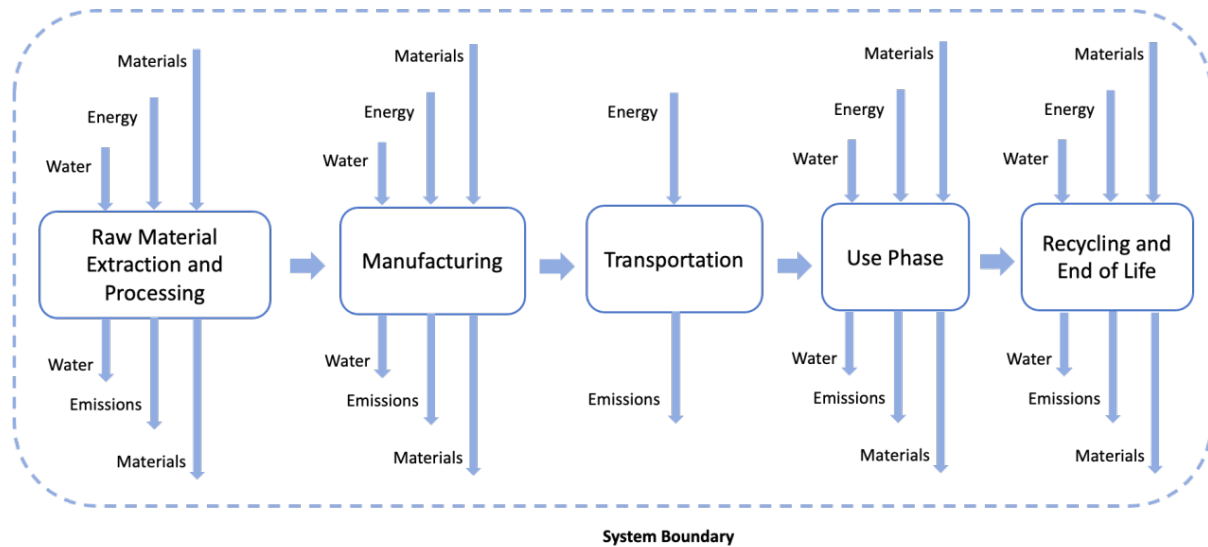


Figure 3.2: System Boundary for LCA Study

3.3. Functional Unit

LCA study on BESS and HESS can use a wide array of different functional units, shown as Table 3.3. Since the purpose of this study is to compare BESS and HESS for connecting renewable energy to grid, 1 MWh of electricity delivered to the grid throughout the project lifetime is used as the functional unit. All environmental impacts will be assessed in per MWh of electricity delivered to the grid for better comparison.

Table 3.3: LCA Studies Related to BESS and HESS

System Components	Life Cycle Stages Considered	Functional Unit	Application	Source
BESS - Battery				
LFP	Cradle-to-grave	10 kWh capacity	Electric vehicles	(Zackrisson et al., 2010)
LFP, NMC, LMO	Cradle-to-gate; Use stage	1 MWh electricity delivered	Grid energy storage	(Hiremath et al., 2015)
LFP, LTO, NMC, LMO, NCA	Cradle-to-gate; Use stage	1 kg battery; 1 kWh delivered; 1 kWh capacity	Grid energy storage	(Baumann et al., 2017)
LFP, NMC, LMO, NCA	Cradle-to-gate; End-of-life	1 MW reserve capacity for one year	Grid frequency regulation	(Ryan et al., 2018)
LFP, NMC, NCA	Cradle-to-grave	Per 1 kWh energy delivered	Grid; Stand-by Scenario	(Yudhistira et al., 2022)
HESS				
Electrolyzer + Hydrogen Storage	Cradle-to-gate	1 kg of green hydrogen at 30 bars	Sustainable green hydrogen production	(Rangel et al., 2025)
HESS+BESS	Cradle-to-Utilization	1 kWh of electricity delivered	Off-grid energy system	(Gandiglio et al., 2022)
HESS+BESS	Cradle-to-gate; Use stage	1 kWh of electricity delivered	Energy storage for buildings	(Le et al., 2024)
HESS	Cradle-to-grave	Annual wind speed	Grid, large scale	(Yamaki et al., 2024)
HESS-AWE	Cradle-to-grave	5MW capacity	Hydrogen production	(Hoppe & Minke, 2025)
HESS-Electrolyzer	Cradle-to-grave	1 kW stack unit; 1 MJ hydrogen production	Hydrogen production	(Wei et al., 2024)
HESS-PEMFC	Cradle-to-grave	1 kWh of electricity delivered; 1 kW capacity	Grid balancing	(Stropanik et al., 2019)

3.4. Life Cycle Inventory (LCI)

Life cycle inventory involves a systematic inventory of the input and output material and energy flows throughout the system boundary. Since direct and consistent comparative LCA studies of BESS and HESS are currently scarce, collecting data on the production of different components from manufacturers is challenging. To obtain the required background data for input and output flows, Ecoinvent 3.11 database is used, which provides comprehensive international industrial and commercial data on material production, transportation, energy consumption, and other relevant factors (Le et al., 2024). Moreover, the LCI data of key components in BESS and HESS are generated based on data sources from relevant literatures with a conversion calculation of FU.

Table 3.4 lists the manufacturing process and operation process inventory data of BESS. Table 3.5 and Table 3.6 summarize the manufacturing process and operation process inventory data of HESS, respectively. However, the modelling of the recycling and end of life (EoL) phase in OpenLCA is excluded due to limited recycling process data and complex allocation requirements. Instead, we manually calculate recycled key material flows in Excel using simplified recycling rates (Stropnik et al., 2019) (Hoppe & Minke, 2025) (Han et al., 2023) to ensure transparency and align with available literature values.

Table 3.4: Inventory Table of BESS (Han et al., 2023)

Stage	Item/Material	Unit	Value/FU
Manufacturing	Water (barrage)	kg	2.54E+00
	Water (cooling)	m ³	3.75E-01
	Dimethyl carbonate	kg	1.77E-01
	Aluminium sheet	kg	1.11E-01
	Copper sheet	kg	1.65E-01

	Ethylene carbonate	kg	1.07E-01
	Carbon black	kg	1.16E-02
	Graphite, battery grade	kg	1.91E-01
	Carboxymethyl cellulose	kg	3.30E-03
	Acrylonitrile-butadiene-styrene	kg	9.57E-02
	Lithium hexafluorophosphate	kg	6.27E-02
	N-methyl-2-pyrrolidone	kg	1.98E-02
	Polypropylene	kg	3.30E-03
	Polyethylene	kg	3.30E-03
	BMS (Battery Management System)	kg	2.61E-02
	Cell container	kg	1.68E-01
	Lithium iron phosphate	kg	4.41E-01
	Polyvinylidene fluoride	kg	1.98E-02
	Electricity (CN mix)	kWh	5.86E+00
	Natural gas	m ³	2.76E-01
Operation and Use	Electricity input - charging	MWh	1.69E+00
	Electricity input - BoP	MWh	3.62E-02
	Electricity delivered to grid	MWh	1.00E+00
End-of-Life	End-of-life input – used BESS	kWh	1.65E-01
	Recovered lithium carbonate	kg	9.66E-02
	Recovered iron phosphate	kg	3.86E-01
	Recovered graphite	kg	5.89E-02

Table 3.5: Manufacturing Process Inventory Table of HESS

System	Component	Material	Unit	Value/FU
Electrolyzer – AWE (Hoppe & Minke, 2025; Wei et al., 2024)				
Stack	Frame	Unalloyed steel	kg	1.24E+00
		Copper	kg	4.93E-02
		Nikel	kg	5.23E-02
	Anode	Nikel	kg	3.13E-01
	Raney buckel cathode	Nikel	kg	1.04E-01
		Aluminum	kg	1.11E-02
		Carbon monoxide	kg	3.71E-03
	Membrane	Zirconium oxide	kg	2.72E-02
		Polyphenelene sulfide	kg	8.42E-03
		Polysulfones	kg	6.42E-03
		N-Methyl-2-pyrrolidone	kg	3.22E-02
	Gasket	Polytetrafluoroethylene	kg	1.93E-03

		Graphite	kg	1.06E-02
		Acrylonitrile butadiene styrene	kg	3.97E-03
		Aniline	kg	1.21E-03
		Acetic anhydride	kg	1.34E-03
		Terephthalic acid	kg	2.18E-03
		Nitric acid	kg	8.16E-04
		Hydrochloric acid	kg	3.22E-03
		Lubricating oil	kg	1.19E-05
	Energy	Electricity	kWh	2.47E-04
BOP	Heat exchanger and fans	steel	kg	8.46E-04
		nickel	kg	2.97E-04
		cast iron	kg	1.63E-04
		stainless steel	kg	1.26E-04
		copper	kg	3.71E-05
		EPDM rubber	kg	1.48E-06
	KOH tank	polyethylene	kg	2.69E-03
	Gas separator	stainless steel	kg	2.80E-03
	Water pump	cast iron	kg	4.40E-03
		stainless steel	kg	3.38E-03
		copper	kg	9.20E-04
	Inverter	steel	kg	1.17E-01
		Lubricating oil	kg	7.19E-02
		copper	kg	2.73E-02
		aluminium	kg	1.07E-02
	Control unit and cable	steel	kg	2.05E-03
		aluminium	kg	1.55E-03
		polyvinylfluoride	kg	6.16E-04
		polyethylene	kg	1.09E-03
		copper	kg	2.30E-04
	Energy	Electricity	kWh	3.19E-05
Hydrogen Storage System (Rossi et al., 2019)				
H ₂ Storage System	Type III H ₂ tank	carbon fibre - liquid ammonia	kg	4.11E-02
		carbon fibre - propylene	kg	1.03E-01
		chromium steel pipe	kg	7.75E-03
		glass fibre reinforced plastic, polyester resin, hand lay-up	kg	1.18E-02
		polyethylene, high density, granulate	kg	2.21E-02
		polymer foaming	kg	1.01E-02
		silicon, electronics grade	kg	1.94E-03
		steel, low-alloyed	kg	2.81E-02
	Energy	Electricity	kWh	2.18E-01
Fuel Cell – PEMFC (Stropnik et al., 2019)				
Stack	Plates, gaskets, membrane, electrodes, catalysts, etc.	Graphite	kg	1.54E-01
		Polyvinylidene chloride (PVdC)	kg	3.77E-02
		Aluminium	kg	1.03E-02
		Chromium steel	kg	3.42E-03

		Glass fibres	kg	3.42E-03
		Perfluorosulphonic acid (PFSA)(Nafion®) – other	kg	2.37E-03
		Perfluorosulphonic acid (PFSA)(Nafion®) – Pt	kg	2.65E-05
		Carbon black	kg	2.74E-05
		Platinum	kg	2.57E-05
BOP	Pipes, tubes, pressure vessels, containers, etc.	Steel product	kg	1.27E-01
		Ployethylene high density granulate (HDPE)	kg	5.14E-02
		Chromium steel	kg	3.77E-02
		Cast-iron component	kg	2.74E-02
		Aluminium	kg	2.57E-02
		Polypropylene granulate (PP)	kg	8.56E-03
Manufacturing and Assembly	Energy	Electricity	kWh	5.79E-01

Table 3.6: Operation Process Inventory Table of HESS

Process	Flow type	Flow	Unit	Value/FU
Electrolyzer – AWE (Wei et al., 2024)				
H ₂ Production	Input	H ₂ O	kg	12896.00
		Electricity to stack	kWh	2251.60
		Electricity to BoP	kWh	157.04
	Output	Waste water	kg	15.44
		O ₂	kg	265.72
		Non-usable heat released	kWh	962.00
		H ₂	kg	43.33
Hydrogen Storage System (Monterey Gardiner, 2009)				
H ₂ Storage	Input	Electricity	kWh	45.50
Fuel Cell – PEMFC (Stropnik et al., 2022)				
Electricity Generation	Input	H ₂	kg	42.30
		Air	kg	2876.55
	Output	Electricity	MWh	1.00
		Water	kg	376.28
		Used air	kg	2542.57
		Heat	kWh	1500.00

3.5. Life Cycle Impact Assessment

To assess the environmental impacts of BESS and HESS, three impact assessment methods in OpenLCA are employed to quantify environmental impacts across various

categories: ReCiPe 2016 v1.03 Midpoint (H), Cumulative Energy Demand (CED), and Environmental Prices.

3.5.1. ReCiPe 2016 v1.03 Midpoint (H)

The ReCiPe 2016 v1.03 Midpoint (H) method provides a midpoint-oriented assessment approach that quantifies the potential environmental impacts of a system by evaluating a wide array of midpoint categories (*RIVM, 2024*). For this study, the focus is placed on two main categories: climate change and material resource use.

- **Climate Change:** Impacts are measured in terms of kilograms of CO₂-equivalents (kg CO₂-eq), accounting for emissions of greenhouse gases such as CO₂, CH₄, and N₂O. These are converted using global warming potential (GWP) factors, enabling a standardized comparison of systems (*Hedgehog, 2024*).
- **Material Resource Use:** Material consumption is characterized in kilograms of copper-equivalents (kg Cu-eq), allowing a unified representation of the depletion potential of raw materials, particularly critical and strategic materials relevant to EU supply chains (*Huijbregts et al., 2017*).

3.5.2. Cumulative Energy Demand (CED)

The Cumulative Energy Demand method quantifies the total primary energy used throughout the life cycle of a product or system, expressed in megajoules equivalent (MJ-eq). It distinguishes between renewable and non-renewable energy sources and is particularly relevant for systems with energy-intensive manufacturing or operational phases (*Röhrlich et al., 2000*). It is especially useful when comparing systems that involve complex conversion and transformation processes, as it is the case with BESS and HESS (*Rangel et al., 2025*).

In this study, CED provides insight into the energy requirements of both BESS and HESS across the life cycle stages, including material extraction, manufacturing, and operation.

3.5.3. Environmental Prices

The Environmental Prices method assigns a monetary value (€/unit) to each environmental impact category based on estimated societal damage costs (Ecochain, 2025). This approach enables the integration of environmental costs into economic decision-making by translating environmental burdens into monetary values.

The total environmental cost for each system is calculated by multiplying the impact results by their respective unit prices:

$$\text{Environmental Cost (€)} = \text{Impact Result} \times \text{Unit Price (€/unit)}$$

The unit price is referred to CE Delf Environmental Prices Handbook EU28 Version (CE Delft, 2020), shown in Table 3.7. The impact result is obtained through the Environmental Prices method in OpenLCA.

Table 3.7: Environmental Prices for Midpoints

Midpoint	Unit	Lower	Central	Upper
Climate change	€/kg CO ₂ -eq.	€ 0.0218	€ 0.0566	€ 0.0944
Ozone depletion	€/kg CFC-eq.	€ 22.1	€ 30.4	€ 45.7
Human toxicity	€/kg 1,4 DB-eq.	€ 0.0725	€ 0.0991	€ 0.153
Smog formation	€/kg NMVOC-eq.	€ 0.84	€ 1.15	€ 1.84
Particulate matter formation	€/kg PM ₁₀ -eq.	€ 28	€ 39.2	€ 60.4
Ionizing radiation	€/kg kBq U235-eq.	€ 0.0297	€ 0.0461	€ 0.0598
Acidification	€/kg SO ₂ -eq.	€ 0.526	€ 4.97	€ 5.66
Freshwater eutrophication	€/kg P-eq.	€ 0.25	€ 1.86	€ 2.11
Marine eutrophication	€/kg N	€ 3.11	€ 3.11	€ 3.11
Land use	€/m ² a	€ 0.0255	€ 0.0845	€ 0.685
Terrrestrial ecotoxicity	€/kg 1,4 DB-eq.	€ 1.17	€ 8.69	€ 9.85
Freshwater ecotoxicity	€/kg 1,4 DB-eq.	€ 0.00485	€ 0.0361	€ 0.0409
Marine ecotoxicity	€/kg 1,4 DB-eq.	€ 0.000992	€ 0.00739	€ 0.00837

3.5.4. Levelized Cost of Storage (LCOS)

The Levelized Cost of Storage (LCOS) is a key techno-economic metric used to assess the cost-effectiveness of energy storage systems over their operational lifespan. It expresses the discounted cost per unit of electricity discharged from the storage system, enabling the comparison of different storage technologies under consistent economic and technical assumptions. Therefore, LCOS is applied in this study to compare the cost-effectiveness of BESS and HESS.

LCOS is calculated using the following equation (Jülch, 2016):

$$LCOS = \frac{CAPEX + \sum_{t=1}^{t=n} \frac{C_t}{(1+r)^t}}{\sum_{t=1}^{t=n} \frac{E_t^{discharged}}{(1+r)^t}}$$

Where:

- CAPEX = capital expenditure of the storage system.
- 1. C_t = annual cost of the storage system in year t, composed of operation cost, reinvestment in storage system components, cost of electricity supply.
- 2. $E_t^{discharged}$ = annual electricity discharged from the storage system in year t.
- r = discount rate.
- n = lifetime of the project (years).

Table 3.7 presents the economic data of both systems, and the technical data refers to Table 3.1 and Table 3.2. Since the scenario is to store renewable energy for grid application, the Levelized Cost of Energy (LCOE) for renewable energy in Belgium is used as the cost of electricity supply. According to the data from IEA (IEA, 2025), the average LCOE of renewable energy projects in Belgium is around 66.2338 €/MWh, with an exchange rate of USD to EUR at 0.89.

Table 3.7: Economic Data of BESS and HESS

System		CAPEX	Unit	OPEX	Unit
BESS (Salvatori Maldonado, 2023)		328091.38	€/MWh	1990.97	€/MWh/year
HESS	AWE (EHO,2024)	1666000	€/MW	43000	€/MW/year
	H ₂ Storage (Dogliani, 2022)	15484	€/m ³	23.23	€/m3/year
	PEMFC (Ahang et al., 2025)	3200000	€/MW	68000	€/MW/year

4. Results and Discussions

In this chapter, a comprehensive LCA comparing BESS and HESS is presented, analyzing their impacts across multiple dimensions. Through a systematic breakdown of climate change contributions, material resource demands, energy use, environmental price, end-of-life recyclability and economic feasibility, this study provides insights into the sustainability trade-offs between these energy storage technologies.

4.1. Climate Change Impact Analysis

This section presents the climate change impacts associated with BESS and HESS, quantified using the ReCiPe 2016 v1.03 Midpoint (H) method. The results are expressed in kilograms of CO₂-equivalents (kg CO₂-eq) per functional unit (1 MWh of electricity delivered to the grid), capturing the global warming potential (GWP) of various greenhouse gas emissions.

The total climate impact for HESS is approximately 137 kg CO₂-eq, nearly 2.5 times that of BESS, which stands at 56 kg CO₂-eq. This significant difference reflects the energy-intensive processes involved in hydrogen production, compression, and storage, in contrast to the more direct electricity-to-battery conversion pathway used in BESS.

As is shown in Figure 4.1, in both systems, fossil-derived CO₂ dominates, accounting for nearly 80 % of the greenhouse-gas weight. HESS releases roughly 109 kg CO₂-eq, compared to about 44 kg CO₂-eq for BESS, mainly coming from electricity, heat and clinker production. The CO₂ Methane (fossil) is the next largest emission, representing 6.5% of HESS's footprint and 7.4% of BESS's, reflecting upstream natural-gas and fugitive emissions. Sulfur hexafluoride (SF₆) and nitrous oxide (N₂O) each account for roughly 5 % of both systems' impacts, while non-fossil methane adds another 2.0 % for HESS and 2.2 % for BESS. The substantially higher CO₂ emissions from HESS can be traced to the energy intensity of hydrogen production, compression, and storage, whereas BESS benefits from a more direct electricity-to-battery conversion and simpler balance-of-plant.

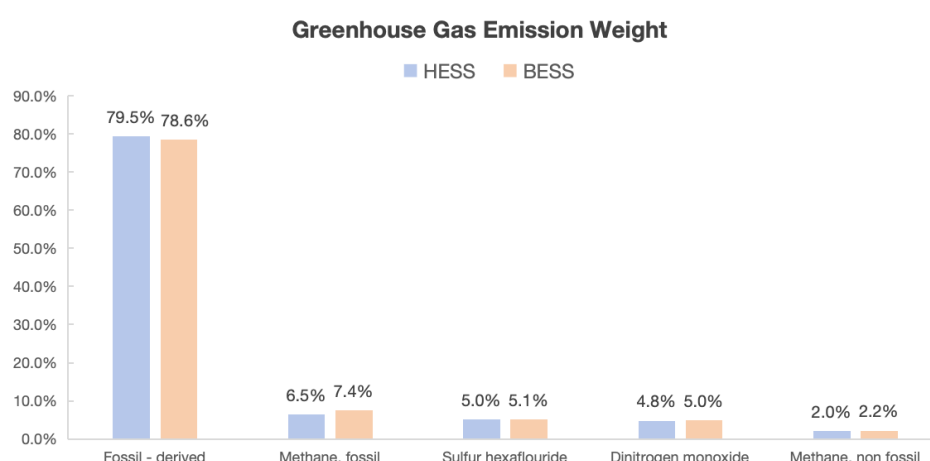


Figure 4.1: Greenhouse Gas Emission Weight of BESS and HESS

The greenhouse gas contribution weight is presented in Figure 4.2. For HESS, roughly 60.9% (83.57 kg) of its greenhouse gas emission comes from the renewable electricity purchased at low voltage and a further 27.0% (37.06 kg) from the low-voltage electricity used in secondary demand, that arises inside the transformation and distribution processes (pumps, controls, auxiliary systems); minor contributors include tap-water supply at 2.9% (3.94 kg) and carbon-fibre components at 2.7% (3.74 kg).

By comparison, BESS's total emission has a 66.3 % (37.11 kg) share from renewable low-voltage electricity, 6.81 % (3.81 kg) from auxiliary systems, 16.6 % (9.3 kg) from synthetic-graphite anode production, and 4.06% (2.27 kg) from lithium iron phosphate production, with all other inputs such as polymers production accounting for 6.2%.

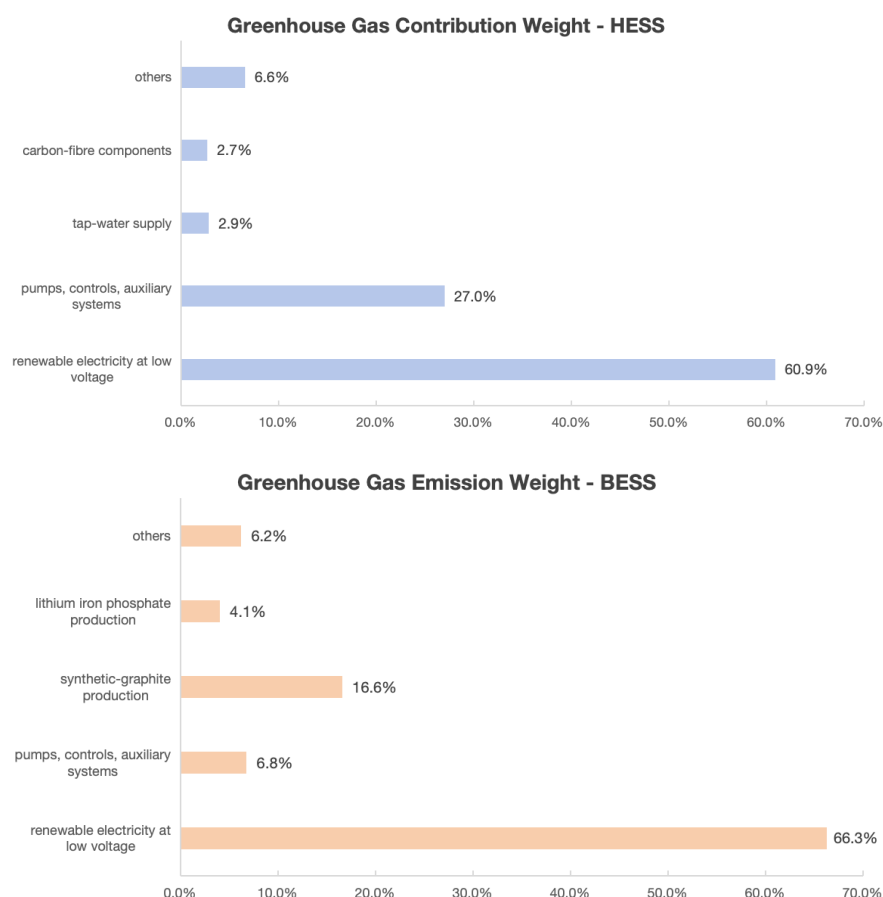


Figure 4.2: Greenhouse Gas Contribution Weight of BESS and HESS

These results underscore the relatively higher carbon footprint of HESS, primarily driven by the complexity and inefficiency of hydrogen's energy conversion chain. In contrast, BESS benefits from fewer conversion stages and lower operational emissions, making it more favorable in terms of climate change mitigation.

4.2. Material Resource Analysis

This section evaluates and compares the use of raw materials BESS and HESS, using the ReCiPe 2016 v1.03 Midpoint (H) method. Material use is expressed in kilograms

of copper-equivalent (kg Cu-Eq) per functional unit, which allows the aggregation of diverse materials based on their relative depletion potential. This analysis helps to reveal whether these materials are critical materials, which would put the supply chain at risk in Europe. The materials used in both BESS and HESS have been categorized in the following Table 4.1, depending on if the materials are:

- **Strategic raw materials:** the EU considers these indispensable for the green-tech, digital, aerospace & defense value chains and sets specific extraction, processing, and recycling targets for them by 2030 (Patel-Campbell, 2023).
 - **Critical raw materials:** economically important and high supply-risk, so they receive monitoring and risk-mitigation measures. All strategic materials are automatically critical, but not vice-versa (EU, 2025).
3. **Non-critical:** the substance is not on the current EU lists. That may change at the 2027 review if market conditions or supply risks evolve.

Table 4.1: Category of Materials Used

Strategic raw materials	Critical (but not strategic)	Currently non-critical
Boron, Cerium, Cobalt, Copper, Gallium, Graphite, Lithium, Magnesium, Manganese, Nickel, Palladium, Platinum, Rhodium, Silicon, Spodumene, Titanium	Aluminum, Antimony, Arsenic, Feldspar, Hafnium, Niobium, Phosphorus, Strontium, Tantalum, Vanadium	Basalt, Cadmium, Chromium, Chrysotile Clay, bentonite Clay, unspecified Diatomite, Gold, Gypsum, Indium, Iodine, Iron, Lead, Magnesite, Mercury, Molybdenum, Perlite, Rhenium, Selenium, Silver, Talc, Tellurium, Tin, Uranium, Zinc

The analysis indicates that both BESS and HESS use the same 51 materials, but the amounts differs significantly, reflecting the distinct components and technologies involved. The results are presented in Figure 4.3.

It shows that BESS consumes 10.74 kg Cu-Eq of materials in total. The main materials used are spodumene (5.04 kg Cu-Eq), cerium (3.24 kg Cu-Eq), copper (0.80 kg Cu-Eq), molybdenum (0.49 kg Cu-Eq), nickel (0.26 kg Cu-Eq), iron (0.20 kg Cu-Eq), magnesite (0.10 kg Cu-Eq) and lithium (0.09 kg Cu-Eq).

While more materials are used for HESS, as 13.04 kg Cu-Eq of materials are used. The main materials used are cerium (7.36 kg Cu-Eq), copper (1.80 kg Cu-Eq), molybdenum (1.15 kg Cu-Eq), nickel (0.71 kg Cu-Eq), iron (0.47 kg Cu-Eq), platinum (0.19 kg Cu-Eq), magnesite (0.23 kg Cu-Eq), and silver (0.17 kg Cu-Eq).

The comparison shows that BESS consumes far more spodumene than hydrogen storage systems. This is a natural consequence of the fact that spodumene is the main source of lithium for lithium-ion batteries, whereas hydrogen technologies require much less. Another observation is that HESS consumes much more cerium than BESS, as this resource is needed to build the electrolyzer and fuel cells electrodes.

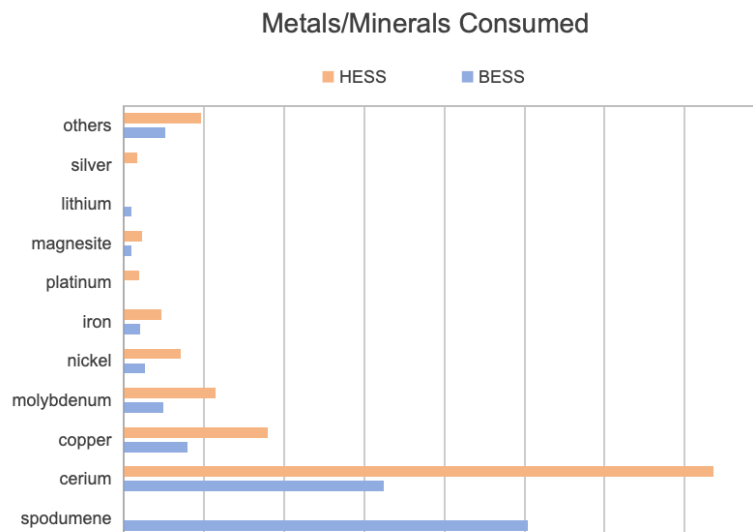


Figure 4.3: Comparison of Material Resources Consumed by BESS and HESS

It is noteworthy that for both systems a significant portion of the material resource demand originates from the infrastructure used to generate renewable electricity for energy storage. Specifically, these upstream contributions account for approximately

37% of the total material use in BESS and 68% in HESS, which matches the energy flow in system configuration that HESS has a higher renewable energy curtailment ratio. This highlights the indirect material burden associated with utilizing renewable energy, particularly in energy storage systems.

Moreover, within BESS, lithium-related materials dominate the overall material profile. Notably, the production of lithium iron phosphate (LFP) contributes 44.5% of the total material demand, while lithium hexafluorophosphate adds another 12%. This underscores the central role of lithium compounds in the environmental footprint of battery storage technologies. The following Figure 4.4 shows the contribution tree for material resource for BESS.

Impact category: Material resources: metals/minerals

Contribution	Process	Required amount	Total result [kg Cu-Eq]	Direct contribution [kg Cu-Eq]
100.00%	BESS LCA - BE	3600.00000 MJ	10.73639	
> 44.48%	lithium iron phosphate production, hydrothermal process lithium iron phosphate C	0.44078 kg	4.77539	
> 36.88%	market for electricity, low voltage, renewable energy products electricity, low volta	6092.30769 MJ	3.95967	
> 19.88%	electricity voltage transformation from medium to low voltage, renewable energy prc	6007.51513 MJ	2.13444	
> 14.45%	market for distribution network, electricity, low voltage distribution network, electri	0.15209 m	1.55149	
> 01.66%	electricity production, photovoltaic, 3kWp slanted-roof installation, multi-Si, panel, r	169.91972 MJ	0.17875	
> 00.88%	electricity production, photovoltaic, 3kWp slanted-roof installation, single-Si, panel,	86.72977 MJ	0.09498	
> 00.00%	market for sulfur hexafluoride, liquid sulfur hexafluoride, liquid Cutoff, U - RER	5.20274E-6 kg	4.38403E-7	
> 12.05%	lithium hexafluorophosphate production lithium hexafluorophosphate Cutoff, U - C	0.06273 kg	1.29407	
> 07.91%	market for hydrogen fluoride hydrogen fluoride Cutoff, U - RoW	0.05313 kg	0.84926	
> 03.90%	market for lithium chloride lithium chloride Cutoff, U - GLO	0.01759 kg	0.41678	
> 00.22%	market for phosphorus pentachloride phosphorus pentachloride Cutoff, U - GLO	0.09189 kg	0.02410	
> 00.01%	market for hydrated lime, packed hydrated lime, packed Cutoff, U - RoW	0.09256 kg	0.00138	
> 00.00%	market for chemical factory, organics chemical factory, organics Cutoff, U - GLO	2.50931E-11 Item(s)	0.00040	
> 00.00%	market group for electricity, medium voltage electricity, medium voltage Cutoff, U	0.09012 MJ	9.18846E-5	
> 00.00%	market for heat, district or industrial, natural gas heat, district or industrial, natural	0.11835 MJ	4.28673E-5	
> 00.00%	market for heat, from steam, in chemical industry heat, from steam, in chemical ind	0.01304 MJ	9.28118E-6	
> 00.00%	market for wastewater, average wastewater, average Cutoff, U - RoW	4.50138E-5 m3	6.60213E-7	
> 02.94%	electrolyte, copper-rich to generic market for copper-rich materials copper-rich ma	0.16509 kg	0.31517	
> 01.48%	electricity production, photovoltaic, 3kWp slanted-roof installation, single-Si, panel,	151.25184 MJ	0.15921	
> 01.30%	steel production, converter, unalloyed steel, unalloyed Cutoff, U - RER	0.11061 kg	0.13922	
> 00.59%	market for synthetic graphite, battery grade synthetic graphite, battery grade Cut	0.19150 kg	0.06302	
> 00.12%	evaporation of natural gas, import from US natural gas, high pressure Cutoff, U - k	0.27569 m3	0.01263	
> 00.08%	dimethyl carbonate production dimethyl carbonate Cutoff, U - RoW	0.17664 kg	0.00818	
> 00.03%	market for ethylene carbonate ethylene carbonate Cutoff, U - GLO	0.10731 kg	0.00364	
> 00.02%	N-methyl-2-pyrrolidone production N-methyl-2-pyrrolidone Cutoff, U - RoW	0.01981 kg	0.00185	
> 00.01%	carboxymethyl cellulose production, powder carboxymethyl cellulose, powder Cut	0.00330 kg	0.00149	
> 00.01%	acrylonitrile-butadiene-styrene copolymer production acrylonitrile-butadiene-styre	0.09575 kg	0.00125	0.00020
> 00.01%	chemical production, organic chemical, organic Cutoff, U - GLO	0.01981 kg	0.00095	
> 00.00%	market for carbon black carbon black Cutoff, U - GLO	0.01156 kg	0.00039	
> 00.00%	polypropylene production, granulate polypropylene, granulate Cutoff, U - RoW	0.00330 kg	0.00018	
> 00.00%	polyethylene, high density, granulate, recycled to generic market for polyethylene, h	0.00330 kg	8.20974E-5	

Figure 4.4: Material Resource Contribution Tree for BESS

To summarize, both technologies rely heavily on strategic and critical minerals, some are so essential that they account for most of the material demand when expressed in kg Cu-eq. Securing a resilient, uninterrupted supply chain for these minerals is, therefore, crucial to the successful roll-out of the projects.

4.3. Energy Use Analysis

Analyzing energy consumption throughout the life cycle is a better indicator for assessing the sustainability of energy storage systems (Han et al., 2023). Previous research has shown that BESS systems require a large amount of energy during the production phase. This is due to the high-intensity processes required to manufacture lithium-ion cells and to extract key materials such as cobalt and nickel (Yudhistira et al., 2022). However, energy consumption is reduced while operating, thus making these solutions good substitutes for short-term energy storage. On the other hand, HESS systems consume more energy while operating, owing to converting electricity to hydrogen and vice versa through fuel cells (Zhao et al., 2020).

This section uses CED method to assess the energy consumption of BESS and HESS. The results are expressed in megajoules equivalent (MJ-eq) per functional unit, with a distinction between renewable and non-renewable sources, shown in Table 4.2.

In terms of total energy demand, HESS shows a total primary-energy requirement of about 19,109 MJ-eq per functional unit, which is about 2.4 times of the BESS's one of 7,924 MJ-eq. This is largely due to the many energy conversions required in HESS, beyond the ongoing operation of auxiliary equipment such as compressors, pumps, and control units, something that has also been observed in previous research (Zhao et al., 2020). As demonstrated in Table 1.2 for the current situation, HESS systems tend to be more energy-intensive compared to other storage technologies, precisely due to the use of critical metals and transformation processes accompanied by efficiency losses (Yudhistira et al., 2022).

In each system, the dominant resource of energy is hydropower, which accounts for about 77% of HESS demand (14,727 MJ-eq) and 83% of BESS demand (6,557 MJ-eq). For BESS, minor energy contributors are fossil fuels, with a 7.4% share (588.5

MJ-eq), followed by solar. HESS, instead, relies more on non-renewable energy than BESS, including nuclear (1,479.9 MJ-eq, 7.7 %) and fossil (1,393.0 MJ-eq, 7.3 %). The different energy consumption structures are also associated with both the Belgian power mix and manufacturing processes required to yield essential components. These findings indicate that BESS is more energy-efficient across its life cycle, especially for applications where minimizing energy consumption is critical. HESS, while offering longer-duration storage potential, entails greater energy overheads that need to be justified by system-level flexibility and decarbonization benefits.

Table 4.2: Energy Demand of BESS and HESS

System	Energy Source	Value (MJ-eq)	Percentage
BESS	Total Renewable Energy	7285.5	91.938%
	Hydraulic energy (potential)	6557.1	82.746%
	Solar Energy (converted)	436.3	5.506%
	Wind Energy (kinetic)	187.9	2.371%
	Biomass (calorific value)	102.5	1.293%
	Geothermal	1.7	0.021%
	Total Non-Renewable Energy	638.84	8.062%
	Fossil	588.52	7.427%
	Nuclear	50.27	0.634%
	Non-renewable biomass	0.05	0.001%
HESS	Total Renewable Energy	16236.08	84.965%
	Hydraulic energy (potential)	14727.24	77.069%
	Solar Energy (converted)	704.62	3.687%
	Wind Energy (kinetic)	530.25	2.775%
	Biomass (calorific value)	269.54	1.411%
	Geothermal	4.43	0.023%
	Total Non-Renewable	2873.05	15.035%
	Nuclear	1479.89	7.744%
	Fossil	1393.04	7.290%
	Non-renewable biomass	0.12	0.001%

4.4. End of Life Recycling Analysis

Table 4.3 presents the end-of-life material recycling analysis for BESS. The results reveal that 79% of the total mass in BESS is theoretically recyclable based on literature

derived recovery rates, primarily from metals such as Aluminum, Copper, and Steel. Lithium carbonate recovery rate reaches 88% while graphite has a recovery rate of 31%, which indicates the challenges due to purification difficulties. The total recycled material is around 0.835 kg/MWh delivered electricity.

Table 4.3: End-of-Life Material Recycling of BESS, FU Based

Materials	Unit	Material Used	Material Recycled	Recycling Rate
Aluminium	kg	0.110608	0.106183	96%
Carbon Black	kg	0.011556	0.008783	76%
Copper	kg	0.178128	0.163878	92%
Graphite	kg	0.191500	0.058936	31%
Iron phosphate	kg	0.440780	0.386301	88%
Lithium carbonate	kg	0.110195	0.096575	88%
Polyethylene	kg	0.003302	0.002773	84%
Polypropylene	kg	0.003302	0.002773	84%
Steel	kg	0.010433	0.009181	88%
Sum	kg	1.059803	0.835384	79%

The result of end-of-life material recycling in HESS is shown in Table 4.4. It shows 89% mass recyclability, dominated by metals like Aluminum, Copper and Nickel. The recovery rate of Platinum is 76%, which indicates the technical achievability of recycling this key expensive metal resource for PEMFC. The total recycled material of HESS is around 2.125 kg/MWh delivered electricity.

Table 4.4: End-of-Life Material Recycling of HESS, FU Based

Materials	Unit	Material Used	Material Recycled	Recycling Rate
Aluminium	kg	0.059332	0.056959	96%
Carbon Black	kg	0.000027	0.000021	76%
Cast Iron	kg	0.031961	0.028125	88%
Copper	kg	0.077874	0.071644	92%
Nikel	kg	0.469989	0.432389	92%
Platinum	kg	0.000052	0.000040	76%
Polyethylene	kg	0.077240	0.064881	84%
Polypropylene	kg	0.111275	0.093471	84%
Steel	kg	1.565671	1.377790	88%
Sum	kg	2.393420	2.125321	89%

This simplified Excel analysis reveals that HESS has slightly higher recyclability due to greater metal content while both systems have the potential to improve its life cycle performance by end-of-life recycling. Though less sophisticated than OpenLCA, it effectively provides actionable insights for material focused sustainability improvements.

4.5. Economic Analysis

4.5.1. Environmental Prices Analysis

To provide an integrated perspective on the environmental impacts of BESS and HESS, this section translates midpoint-level results into monetary terms using the Environmental Prices method. By assigning unit costs to each environmental impact category, the analysis estimates the total societal cost in euros per functional unit.

Table 4.5 and 4.6 show the calculated environmental costs for both BESS and HESS. The total environmental cost for BESS is 30.39 €/MWh, while for HESS is 75.31 €/MWh. This represents around 148% higher environmental cost for HESS than BESS, reflecting its greater overall environmental burden across multiple categories.

Moreover, HESS scores higher in nearly every impact category, especially toxic emissions and particulate matter, which can be attributed to upstream activities such as component manufacturing and complex energy transformation steps.

These monetary impact values provide actionable insight for decision-making. BESS is more environmentally cost-efficient, with lower external costs per MWh of electricity stored and supplied. HESS, despite its potential for seasonal storage and long-term balancing, imposes a significantly higher societal burden.

Table 4.5: Environmental Costs for BESS, FU Based

Impact category	Unit	Impact Value	Unit price (€)	Cost (€)
Climate change	kg CO ₂ eq	54.40536	0.0566	3.08
Ozone depletion	kg CFC-11 eq	0.000002	30.4	0.00
Human toxicity	kg 1,4-DB eq	116.5892	0.0991	11.55
Photochemical oxidant formation	kg NMVOC eq	0.281146	1.15	0.32
Particulate matter formation	kg PM ₁₀ eq	0.237316	39.2	9.30
Ionizing radiation	kBq U ₂₃₅ eq	3.203257	0.0461	0.15
Terrestrial acidification	kg SO ₂ eq	0.576954	4.97	2.87
Freshwater eutrophication	kg P eq	0.05318	1.86	0.10
Marine eutrophication	kg N eq	0.020784	3.11	0.06
Agricultural land occupation	m ² a	10.27496	0.0845	0.87
Terrestrial ecotoxicity	kg 1,4-DB eq	0.028445	8.69	0.25
Freshwater ecotoxicity	kg 1,4-DB eq	43.11862	0.0361	1.56
Marine ecotoxicity	kg 1,4-DB eq	37.33229	0.00739	0.28
Total				30.39

Table 4.6: Environmental Costs for HESS, FU Based

Impact category	Unit	Impact Value	Unit price (€)	Cost (€)
Climate change	kg CO ₂ eq	133.715952	0.0566	7.57
Ozone depletion	kg CFC-11 eq	0.000014	30.4	0.00
Human toxicity	kg 1,4-DB eq	279.607252	0.0991	27.71
Photochemical oxidant formation	kg NMVOC eq	0.652599	1.15	0.75
Particulate matter formation	kg PM ₁₀ eq	0.539843	39.2	21.16
Ionizing radiation	kBq U ₂₃₅ eq	81.308674	0.0461	3.75
Terrestrial acidification	kg SO ₂ eq	1.379323	4.97	6.86
Freshwater eutrophication	kg P eq	0.129940	1.86	0.24
Marine eutrophication	kg N eq	0.043635	3.11	0.14
Agricultural land occupation	m ² a	28.089485	0.0845	2.37
Terrestrial ecotoxicity	kg 1,4-DB eq	0.052456	8.69	0.46
Freshwater ecotoxicity	kg 1,4-DB eq	101.380382	0.0361	3.66
Marine ecotoxicity	kg 1,4-DB eq	87.773069	0.00739	0.65
Total				75.31

4.5.2. LCOS Analysis

Based on the methodology presented in section 3.5, the calculated LCOS of BESS is around 116.11 €/MWh, which is comparable to industrial reference range of 110.36 to 263.44 €/MWh (Colthorpe, 2024). The LCOS of HESS is calculated as 259.81 €/MWh, which is also within the industrial reference from 231.00 to 590.00 €/MWh (Fraunhofer ISE, 2024). The LCOS of HESS is much more expensive compared to BESS, primarily due to the higher capital and operational costs associated with integrating hydrogen systems. This includes additional components like electrolyzers, hydrogen storage tanks, and fuel cells, all of which significantly increase system complexity, maintenance requirements, and overall cost per unit of energy delivered. Furthermore, the overall efficiency of HESS (27.27%) is typically lower than that of BESS (59.09%), resulting in less energy output per unit of input, further driving up the levelized cost.

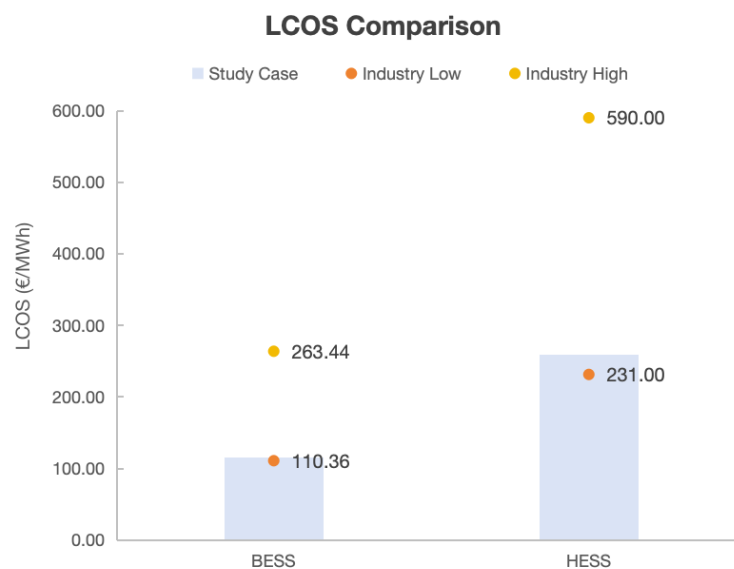


Figure 4.5: LCOS Comparison of BESS and HESS

The sensitivity analysis is conducted to evaluate the impact of key parameters on LCOS for both BESS and HESS. Each parameter is varied by $\pm 10\%$ to assess its influence on LCOS, shown in Figure 4.6 and Figure 4.7.

For BESS, electricity cost exhibits the highest sensitivity, with a $\pm 9.6538\%$ change in LCOS for a $\pm 10\%$ variation in electricity cost. This underscores the critical role of electricity pricing in the economic viability of BESS. CAPEX shows relatively moderate sensitivity, with a $\pm 0.3321\%$ change in LCOS.

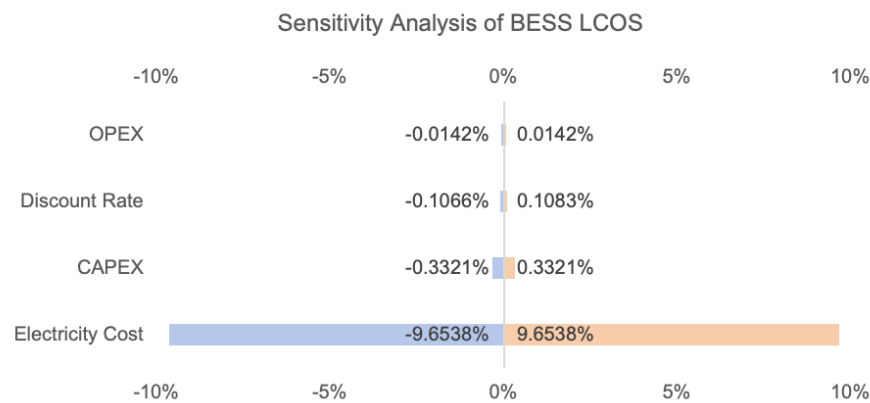


Figure 4.6: Sensitivity Analysis of BESS LCOS

For HESS, the situation is similar. Again, electricity cost is the most influential parameter, with a $\pm 9.3475\%$ change in LCOS for a $\pm 10\%$ variation. CAPEX of PEMFC shows notable sensitivity, with a $\pm 0.3003\%$ change in LCOS, reflecting the cost sensitivity of PEMFC. CAPEX of AWE also shows meaningful contribution, with a $\pm 0.2246\%$ change in LCOS.

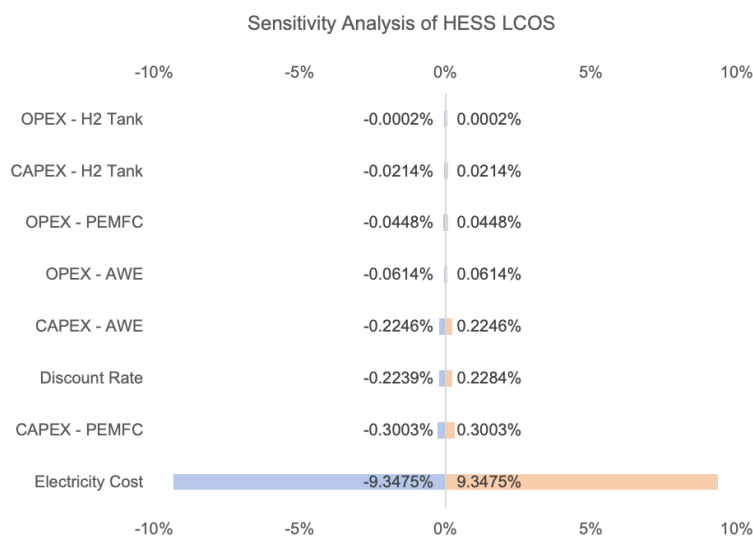


Figure 4.7: Sensitivity Analysis of HESS LCOS

In conclusion, the sensitivity analysis reveals that electricity cost is the most critical driver of LCOS for both BESS and HESS. CAPEX for major components, such as PEMFC, AWE and general BESS infrastructure, also play significant roles. It indicates that efforts to reduce electricity costs and optimize capital investment would yield the highest returns in improving the economic feasibility of both energy storage systems.

4.5.3. Environmental externality-Adjusted LCOS Analysis

To further capture the full societal impact of energy storage systems, environmental externalities is integrated into LCOS to compute the environmental externality-adjusted LCOS. By combining environmental costs and LCOS, the environmental externality-adjusted LCOS reflects not only the financial cost of building and operating the energy storage system, but also the hidden environmental costs typically borne by society. This provides a more holistic and policy-relevant metric for comparing energy storage technologies in the context of sustainability and long-term climate goals.

As is shown in Figure 4.8, the adjusted LCOS for BESS is 146.46 €/MWh, while the adjusted LCOS for HESS is 335.12 €/MWh, which is 2.3 times of BESS. Considering the environmental costs, the gap between both systems has increased from 143.70 €/MWh to 188.63 €/MWh, around 31% increase. This suggests that BESS is not only more economically efficient but also more environmentally favorable under the given assumptions. Therefore, with the internalization of environmental costs, BESS presents a more sustainable and cost-effective solution compared to HESS in this context.

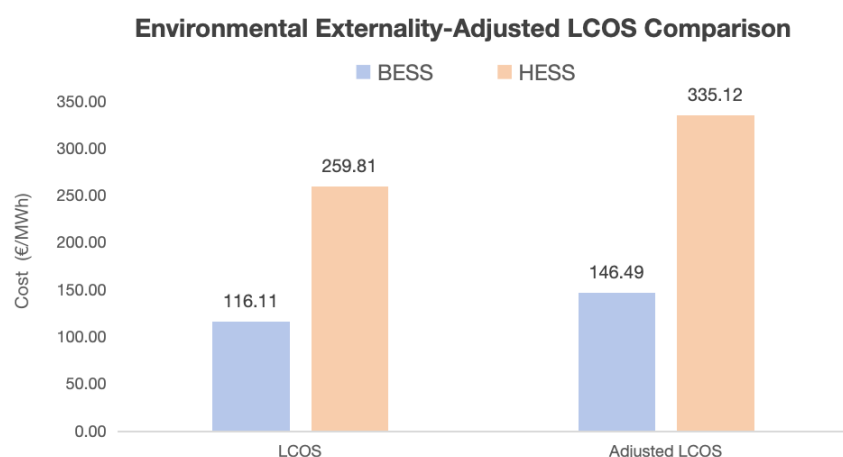


Figure 4.8: Environmental Externality-Adjusted LCOS Comparison

4.6. Sensitivity Analysis

To further assess the robustness of the environmental impacts of BESS and HESS, a sensitivity analysis is conducted based on two critical parameters: renewable energy curtailment ratio and system operation hours, each varied by $\pm 10\%$. This analysis helps to identify which environmental impact indicators are most influenced by these variations and thus where optimization efforts should be prioritized. The results of carbon emission, material resource, energy use and environmental cost for BESS and HESS are shown in Table 4.7 and 4.8. The influence assessed in percentage is presented in Figure 4.9.

Table 4.7: Sensitivity Analysis Results for BESS, FU Based

Impacts	Unit	Baseline	Curtailment Ratio +10%	Curtailment Ratio -10%	Operation Hours +10%	Operation Hours -10%
Climate change	kg CO ₂ Eq	56.00	57.56	54.55	54.58	57.73
Material: metals/minerals	kg Cu-Eq	10.74	10.90	10.58	10.13	11.47
Energy Use	MJ-Eq	7924.34	8237.62	7635.40	7897.29	7957.40
Environmental Cost	€	30.39	31.47	29.39	30.05	30.79

Table 4.8: Sensitivity Analysis Results for HESS, FU Based

Impacts	Unit	Baseline	Curtailment Ratio +10%	Curtailment Ratio -10%	Operation Hours +10%	Operation Hours -10%
Climate change	kg CO ₂ Eq	137.32	143.54	131.91	136.51	138.27
Material: metals/minerals	kg Cu-Eq	13.04	13.70	12.46	12.80	13.31
Energy Use	MJ-Eq	19109.13	20354.44	18025.09	19097.41	19123.01
Environmental Cost	€	75.31	79.60	71.57	74.85	75.86

In terms of energy curtailment ratio, increased curtailment leads to higher environmental burdens per FU for both systems, while reducing curtailment has the opposite effect. For BESS, climate change impact increases by 2.79% and energy use by 3.95% under a 10% rise in curtailment. The environmental cost also increases by 3.56%, while material resources only increases by 1.55%. In contrast, HESS shows much greater sensitivity. A 10% increase in curtailment results in +6.52% in energy use and +5.70% in environmental cost. This reflects the higher energy and conversion losses inherent in HESS. These results highlight that curtailment management is particularly critical for HESS, as it exacerbates inefficiencies more significantly than in BESS.

Operation hours influence the utilization rate of energy storage infrastructure, thus affecting the capacity design of the system. BESS shows strong sensitivity in material use that decreasing operation time by 10% leads to a 6.87% increase in material footprint whereas increasing operation time reduces it by 5.62%. Carbon emissions vary moderately, with a 10% decrease in operation time resulting in a 3.10% increase in carbon emissions. Energy use is the least sensitive with less than 1% variation. HESS, however, is relatively insensitive to changes in daily operation duration. All four impact indicators vary by less than $\pm 2\%$, with energy use nearly unaffected. This

indicates that BESS performance is more closely tied to utilization rate, while HESS is dominated by its energy conversion characteristics regardless of operational time.

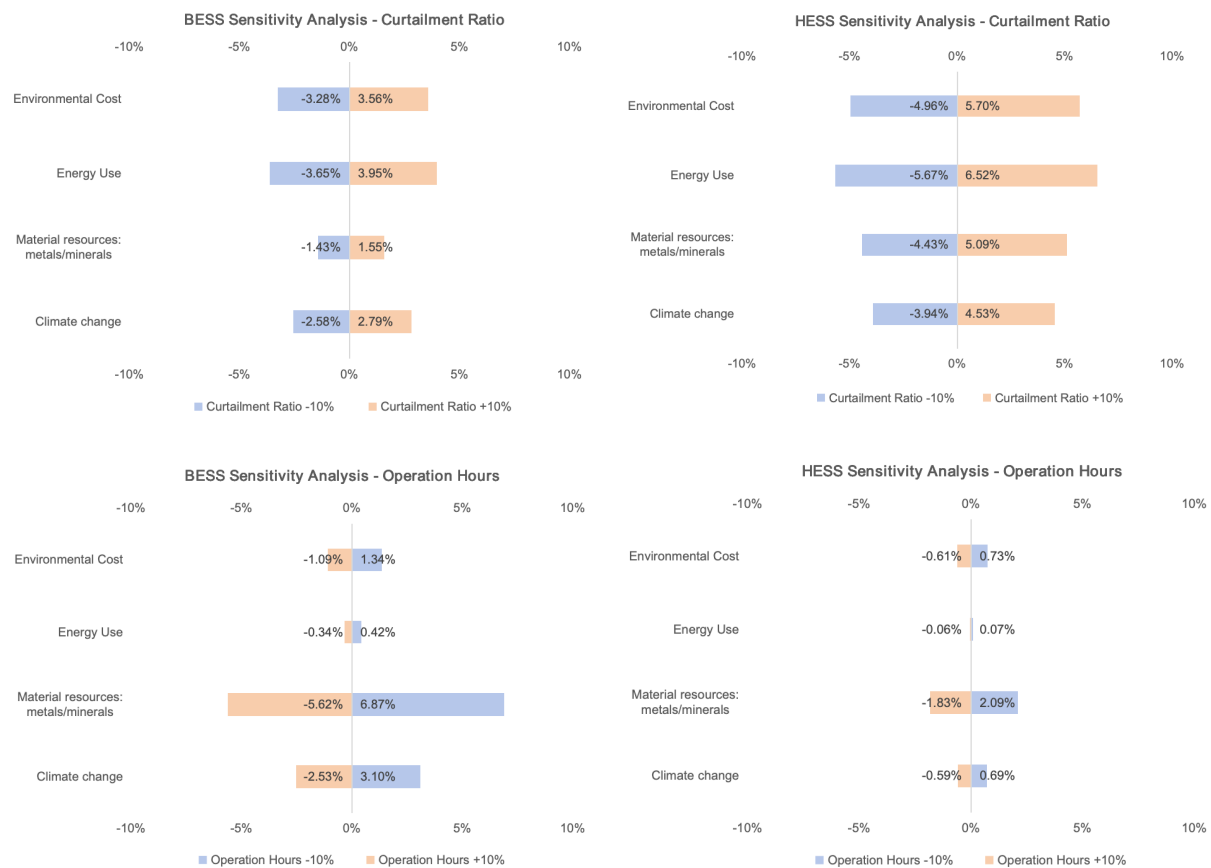


Figure 4.9: Sensitivity Analysis Based on Curtailment Ratio and Operation Hours

5. Conclusions and Recommendations

Through the comparative life cycle assessment (LCA) of BESS and HESS in the context of integrating renewable energy into the grid, it highlights that BESS is a more efficient, cost effective, and environmentally friendly option for short- to medium-term energy storage, while hydrogen energy storage (HESS) plays a key role in large-scale and seasonal applications where longer storage is needed. Although hydrogen shows potential for future energy systems, it currently comes with higher energy losses, greater material use, and a larger environmental footprint. However, it cannot be

denied that both technologies have a place in a renewable energy portfolio and can be complementary when applied to their respective strengths.

This study has therefore many key takeaways and recommendations that can be proposed to guide future development. For instance, policy should promote investment in battery recycling infrastructure and second-life applications, while also advancing the development of cleaner electrolyzers and improving the recovery of critical metals for hydrogen systems. From a technology perspective, these innovations will help reduce the environmental footprint of both battery and hydrogen storage and make their production and disposal more efficient, so a good idea would be to invest in advancing battery recycling infrastructure and second-life applications, and simultaneously, promote cleaner electrolyzer development and improved recycling of critical metals for hydrogen systems. At the research level, there is a clear need for more real-world, system-wide LCA studies that consider operational strategies and end-of-life management to reflect actual conditions and sustainability challenges. Deployment decisions should be made based on application context; batteries are more suitable for short-term, high-frequency storage due to their higher round-trip efficiency and lower cost, while hydrogen is a viable option for long-term, large-scale, or seasonal storage where flexibility and duration are more important than raw efficiency.

By aligning technological advancement, policy design and deployment strategies, both storage technologies can contribute to a more sustainable and resilient energy transition.

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Appendices

Screenshots of OpenLCA Model - BESS

Inputs/Outputs – BESS LCA – BE

▼ Inputs

Flow	Category	Amount	Unit	Costs/Revenue	Uncertainty
acrylonitrile-butadiene-styrene	C:Manufacturing/22:Manufact	0.09575	kg		none
Battery managment system	BESS LCA - BE	0.02608	kg		none
carbon black	C:Manufacturing/20:Manufact	0.01156	kg		none
carboxymethyl cellulose, p	C:Manufacturing/20:Manufact	0.00330	kg		none
Cell container - BE	BESS LCA - BE	0.16839	kg		none
chemical, organic	C:Manufacturing/20:Manufact	0.01981	kg		none
copper-rich materials	C:Manufacturing/24:Manufact	0.16509	kg		none
dimethyl carbonate	C:Manufacturing/20:Manufact	0.17664	kg		none
electricity, low voltage	D:Electricity, gas, steam and e	5.86055	kWh		none
electricity, low voltage	D:Electricity, gas, steam and e	0.03615	MWh		none
electricity, low voltage, rer	D:Electricity, gas, steam and e	1.69231	MWh		none
ethylene carbonate	C:Manufacturing/20:Manufact	0.10731	kg		none
lithium hexafluorophospha	C:Manufacturing/20:Manufact	0.06273	kg		none
lithium iron phosphate	C:Manufacturing/20:Manufact	0.44078	kg		none
N-methyl-2-pyrrolidone	C:Manufacturing/20:Manufact	0.01981	kg		none
natural gas, high pressure	D:Electricity, gas, steam and e	0.27569	m3		none
polyethylene, high density	C:Manufacturing/20:Manufact	0.00330	kg		none
polypropylene, granulate	C:Manufacturing/20:Manufact	0.00330	kg		none
steel, unalloyed	C:Manufacturing/24:Manufact	0.11061	kg		none
synthetic graphite, battery	C:Manufacturing/27:Manufact	0.19150	kg		none
Water, cooling, unspecifie	../Resource/in water	0.37475	m3		none
Water, river	../Emission to water/unspecifi	2.54233	l		none

▼ Outputs

Flow	Category	Amount	Unit	Costs/Revenue	Uncertainty
1 MWh electricity deliver	BESS LCA - BE	1.00000	MWh		none

Screenshots of OpenLCA Model - HESS

Inputs/Outputs - AEW- Hyogren production - All - BE

Inputs

Flow	Category	Amount	Unit	Costs/Rev...	Uncertainty	Avoided w...	Provider	Data quali...	Location	Description
acetic anhydride	C:Manufacturing/20:M...	0.00134	kg		none		market ...			
acrylonitrile-butadien...	C:Manufacturing/22:M...	0.00397	kg		none		market ...			
aluminium alloy, met...	C:Manufacturing/24:M...	0.01070	kg		none		market ...			
aluminium alloy, met...	C:Manufacturing/24:M...	0.01110	kg		none		market ...			
aluminium alloy, met...	C:Manufacturing/24:M...	0.00155	kg		none		market ...			
aluminium alloy, met...	C:Manufacturing/24:M...	0.01030	kg		none		market ...			
aluminium alloy, met...	C:Manufacturing/24:M...	0.02570	kg		none		market ...			
aniline	C:Manufacturing/20:M...	0.00121	kg		none		market ...			
carbon black	C:Manufacturing/20:M...	2.74000E-5	kg		none		market ...			
carbon fibre reinforce...	C:Manufacturing/22:M...	0.10300	kg		none		market ...			
carbon fibre reinforce...	C:Manufacturing/22:M...	0.04110	kg		none		market ...			
carbon monoxide	C:Manufacturing/20:M...	0.00371	kg		none		market ...			
cast iron	C:Manufacturing/24:M...	0.00440	kg		none		market ...			
carbon monoxide	C:Manufacturing/20:M...	0.00371	kg		none		market ...			
cast iron	C:Manufacturing/24:M...	0.00440	kg		none		market ...			
cast iron	C:Manufacturing/24:M...	0.00016	kg		none		market ...			
cast iron	C:Manufacturing/24:M...	0.02740	kg		none		market ...			
chromium steel pipe	C:Manufacturing/24:M...	0.00775	kg		none		market ...			
electricity, low voltage	D:Electricity, gas, steam...	3.19000E-5	kWh		none		market ...			
electricity, low voltage	D:Electricity, gas, steam...	0.00025	kWh		none		market ...			
electricity, low voltage	D:Electricity, gas, steam...	157.04000	kWh		none		market ...			
electricity, low voltage	D:Electricity, gas, steam...	0.57900	kWh		none		market ...			
electricity, low voltage	D:Electricity, gas, steam...	0.21800	kWh		none		market ...			
electricity, low voltage	D:Electricity, gas, steam...	45.50000	kWh		none		market ...			
electricity, low voltag...	D:Electricity, gas, steam...	3563.70000	kWh		none		market ...			
glass fibre reinforced ...	C:Manufacturing/22:M...	0.01180	kg		none		market ...		BE	
glass fibre reinforced ...	C:Manufacturing/22:M...	0.00342	kg		none		market ...			
graphite	B:Mining and quarrying...	0.01060	kg		none		market ...			
graphite	B:Mining and quarrying...	0.15400	kg		none		market ...			
hydrochloric acid, wit...	C:Manufacturing/20:M...	0.00322	kg		none		market ...			
lubricating oil	C:Manufacturing/19:M...	1.19000E-5	kg		none		market ...			
lubricating oil	C:Manufacturing/19:M...	0.07190	kg		none		market ...			
metal working, avera...	C:Manufacturing/25:M...	0.00342	kg		none		market ...			
metal working, avera...	C:Manufacturing/25:M...	0.00023	kg		none		market ...			
metal working, avera...	C:Manufacturing/25:M...	3.71000E-5	kg		none		market ...			
lubricating oil	C:Manufacturing/19:M...	0.07190	kg		none		market ...			
metal working, avera...	C:Manufacturing/25:M...	0.00342	kg		none		market ...			
metal working, avera...	C:Manufacturing/25:M...	0.00023	kg		none		market ...			
metal working, avera...	C:Manufacturing/25:M...	0.02730	kg		none		market ...			
metal working, avera...	C:Manufacturing/25:M...	0.04930	kg		none		market ...			
metal working, avera...	C:Manufacturing/25:M...	0.00092	kg		none		market ...			
metal working, avera...	C:Manufacturing/25:M...	0.12700	kg		none		market ...			
N-methyl-2-pyrrolido...	C:Manufacturing/20:M...	0.03220	kg		none		market ...			
nickel concentrate, 16...	C:Manufacturing/24:M...	0.10400	kg		none		market ...			
nickel concentrate, 16...	C:Manufacturing/24:M...	0.00030	kg		none		market ...			
nickel concentrate, 16...	C:Manufacturing/24:M...	0.31300	kg		none		market ...			
nickel concentrate, 16...	C:Manufacturing/24:M...	0.05230	kg		none		market ...			
nitric acid, without wa...	C:Manufacturing/20:M...	0.00082	kg		none		market ...			
platinum	B:Mining and quarrying...	2.57000E-5	kg		none		market ...			
platinum	B:Mining and quarrying...	2.65000E-5	kg		none		market ...			
polyethylene, high de...	C:Manufacturing/20:M...	0.00269	kg		none		market ...			
polyethylene, high de...	C:Manufacturing/20:M...	0.00109	kg		none		market ...			
polyethylene, high de...	C:Manufacturing/20:M...	0.02210	kg		none		market ...			
polyethylene, high de...	C:Manufacturing/20:M...	0.05140	kg		none		market ...			
polymer foaming	C:Manufacturing/22:M...	0.01010	kg		none		market ...			
polyphenylene sulfide	C:Manufacturing/20:M...	0.00842	kg		none		market ...			

polyphenylene sulfide	C:Manufacturing/20:M...	0.00842	kg		none	market ...			
polypropylene, granu...	C:Manufacturing/20:M...	0.00856	kg		none	market ...			
polysulfone	C:Manufacturing/20:M...	0.00642	kg		none	market ...			
polysulfone	C:Manufacturing/20:M...	0.00237	kg		none	market ...			
polyvinylfluoride	C:Manufacturing/20:M...	0.00062	kg		none	market ...			
polyvinylidenchloride,...	C:Manufacturing/20:M...	0.03770	kg		none	market ...			
purified terephthalic a...	C:Manufacturing/20:M...	0.00218	kg		none	market ...			
silicon, electronics gra...	C:Manufacturing/20:M...	0.00194	kg		none	market ...			
steel, chromium steel ...	C:Manufacturing/24:M...	0.03770	kg		none	market ...			
steel, chromium steel ...	C:Manufacturing/24:M...	0.00338	kg		none	market ...			
steel, chromium steel ...	C:Manufacturing/24:M...	0.00013	kg		none	market ...			
steel, chromium steel ...	C:Manufacturing/24:M...	0.00280	kg		none	market ...			
steel, low-alloyed	C:Manufacturing/24:M...	0.00085	kg		none	market ...			
steel, low-alloyed	C:Manufacturing/24:M...	0.11700	kg		none	market ...			
steel, low-alloyed	C:Manufacturing/24:M...	0.00205	kg		none	market ...			
steel, low-alloyed	C:Manufacturing/24:M...	0.02810	kg		none	market ...			
steel, unalloyed	C:Manufacturing/24:M...	1.24000	kg		none	market ...			
synthetic rubber	C:Manufacturing/20:M...	1.48000E-6	kg		none	market ...			
tap water	E:Water supply; sewera...	1.28960E4	kg		none	market ...			
tetrafluoroethylene	C:Manufacturing/20:M...	0.00193	kg		none	market ...			
zirconium oxide	C:Manufacturing/20:M...	0.02720	kg		none	market ...			
Oxygen	./Resource/in air	2876.55000	kg		none				

Outputs

Flow	Category	Amount	Unit	Costs/Rev...	Uncertainty	Avoided p...	Provider	Data quali...	Location	Description
AEW- Hyogren pro...	L: Project	1.00000	MWh		none					
Carbon dioxide, in air	./Resource/in air	2542.57000	kg		none					
Heat, waste	./Emission to air/low p...	962.00000	kWh		none					
Heat, waste	./Emission to air/low p...	1500.00000	kWh		none					
Oxygen	./Emission to air/unsp...	265.72000	kg		none					
Waste water	./Emission to water/fre...	376.28000	kg		none					
Water	./Emission to air/low p...	0.01544	m3		none					

General information Inputs/Outputs Documentation Parameters Allocation Social aspects Direct impacts